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Assessing the implications of using RDTS in donor blood screening for transfusion-transmissible infections (TTIs): analysis of trans-in donor blood at a Tertiary Hospital in Ghana

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ABSTRACT

Background: Blood transfusion is a critical hemotherapeutic intervention. However, the widespread utilization of Rapid Diagnostic Test (RDT) methods in screening potential donors for TTIs poses a serious public health challenge. We investigated the prevalence of TTIs in donor blood units that were transported from external health facilities (Trans-in blood) to Ho Teaching Hospital for critical care.

Method: We adopted a cross-sectional design to investigate 727 units of Trans-in blood at the Ho Teaching Hospital's blood bank between January 1, 2021, to December 31, 2022. An *apriori-developed* data capture form was used to collect the data for this study. The chi-square test and Cramer's V or Phi were used to assess the association between variables of interest. Binary logistic regression analysis was used to determine the association between TTIs and donor's blood groups whereas spatial visualization was employed to present the distribution of missed TTIs.

Results: The overall prevalence of missed TTIs was 6.19 % (95 % CI: 4.66–8.18), comprising missed HIV, HBV, and Syphilis infection rates of 1.79 % (95 % CI: 1.72–1.86), 1.24 % (95 % CI: 1.17–1.31), and 3.16 % (95 % CI: 3.09–3.23), respectively. There was a higher rate of missed HIV infections in 2022 (2.04 %, 95 % CI: 1.04–3.97) compared to 2021 (1.49 %, 95 % CI: 0.64–3.45), and a decline in the rates of missed HBV and Syphilis infection within the period. The prevalence of TTI was highest among donors with ABO blood group 'O' (68.89 %, 95 % CI: 54.34–80.47) and Rh 'D' Positive blood group (95.56 %, 95 % CI: 85.17–98.77). Compared to donors with blood group O, non-O donors had 4.53 times the odds (95 % CI: 1.12–18.27; p-value=0.034) of missed HBV infection.

Conclusion: The observed prevalence of missed TTIs highlights potential gaps in current screening practices, posing a significant public health concern. Efforts made to clamp down on the spread of these infections within the population may be impeded if the current trend of missed TTIs continues. Further studies on a larger scale with more comprehensive data are advised to better appreciate the public health implications of missed TTIs in Ghana.

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Introduction

Blood transfusion remains an undeniably critical therapeutic intervention in contemporary medicine. However, the potentially life-threatening complications associated with its application pose a significant challenge [1]. The risks of transfusing blood-borne pathogens to persons who are often already grappling with life-threatening conditions such as End-Stage Renal Disease (ESRD), Sickle Cell Anemia (SCA), and trauma from road accidents are considerable [2]. Global reports indicate that over 81 million units of blood are donated annually [3,4]. Furthermore, approximately 16 million and 5 million new infections with Hepatitis B Virus (HBV) and Hepatitis C Virus (HCV), respectively, attributable to the transfusion of contaminated blood, are recorded each year [3]. Indeed, the transfusion of infected blood has been reported to account for 5–10 % of HIV infections in Sub-Saharan Africa and 12.5 % of post-transfusion hepatitis cases underscoring the critical need for stringent screening and safety measures in blood transfusion practices [2,5-8].

Globally, the prevalence of TTIs among blood donors exhibits significant heterogeneity, which is contingent upon the endemic prevalence within respective populations [2]. The Human Immunodeficiency Virus (HIV) epidemic continues to pose a daunting challenge to public health worldwide. According to the World Health Organization (WHO), approximately 37.7 million (30.2–45.1 million) individuals were living with HIV at the end of 2020. Notably, the African region of the WHO remains disproportionately afflicted, with an estimated 3.6 % of adults living with HIV. In 2020 alone, there were 1.5 million (1.0–2.0 million) new infections and about 680,000 HIV-related fatalities globally [9]. Data suggests that 15 % of HIV transmission in Sub-Saharan Africa is attributable to blood transfusion [10]. HIV prevalence rates among blood donors on the African continent have been reported to range from 0.1 % in Ethiopia to as high as 30.77 % in Ghana [11-18].

Concerning transfusion-associated Hepatitis B Virus (HBV) and Hepatitis C Virus (HCV) infections, epidemiological studies have reported a diverse range of findings. According to data from WHO, the global morbidity and mortality rates for HBV and HCV infections in 2019 were 296 million chronic infections and 820,000 deaths for HBV, and 58 million chronic infections and 290,000 deaths for HCV. The WHO also noted that 81 million and 9 million of the global HBV and HCV infections, respectively, were from the WHO African regions, and approximately 1.5 million new infections each for HBV and HCV are recorded each year [19]. At the country level, Nlankpe and colleagues [20] reported HBV and HCV seroprevalence rates of 9.59 % (95 % CI: 9.00–10.20) and 12.71 % (95 % CI: 12.00–13.40), respectively, among blood donors in Saboba District of Ghana. Comparable studies in Tanzania reported seroprevalence rates of 5.1 % for HBV and 1.1 % for HCV [1], while in Northwest Ethiopia, seroprevalence rates of 3.6 % for HBV and 0.5 % for HCV were documented [2].

Transfusion-transmissible infections can be broadly classified into three categories: viral, bacterial, and parasitic infections, with viral infections being the most commonly encountered concerning transfusion. Persistent post-transfusion infections have predominantly been caused by Human Immunodeficiency Virus (HIV) types 1 and 2, Hepatitis B Virus (HBV), Hepatitis C Virus (HCV), and syphilis [21,22]. This remains a significant public health issue, especially in low-resource settings where there is limited availability of highly sensitive screening tests for these pathogens. Consequently, approximately 1.6 million units of blood units are discarded each year worldwide, with 10 % of these originating from sub-Saharan Africa [23]. Most countries, including Ghana, have adopted the World Health Organization's recommendation of subjecting all blood donors and units to laboratory screening for the four mandatory TTIs, namely HIV types 1 and 2, HBV, HCV, and syphilis [24] However, only countries with more robust screening methodologies have succeeded in significantly reducing the incidence and prevalence rates of these TTIs [10,21].

The high risk of TTIs, particularly in low-resource settings, can be attributed to several factors. These include the endemicity of specific TTIs in the general population, the over-reliance on Family Replacement Blood Donors (FRBDs) in contrast to the WHO's recommendation of collecting blood only from Voluntary Non-Remunerated Blood Donors (VNRBDs) from low-risk population groups [25], inadequate screening facilities and surveillance systems, and the absence of reference screening tests like Enzyme Immunoassays (EIA) or Nucleic Acid Testing (NAT) [4,21]. Additionally, the over-reliance on rapid test-based screening kits, especially at district health hospitals, as well as donors in the window period or donors with markers of chronic infection below the sensitivity of existing assays, pose a substantial risk of TTI [21].

As we approach the deadline for the attainment of the Sustainable Development Goals (SDGs), specifically targets 3.3 and 3.8, in less than seven years, the global health community is tasked with the eradication of communicable diseases, inclusive of AIDS and hepatitis, and the assurance of universal access to safe, efficacious, quality, and affordable essential medicines, including blood, by 2030 [26]. Although the demand for blood products in Ghana is perpetually high due to a variety of health issues [27], the country is however grappling with significant challenges, particularly those associated with the accurate detection of TTIs in donor blood, which threaten to undermine the progress made by the Ghana National Blood Service over the years. The extant national blood policy (2nd edition), revised in October 2020 and adopted under the theme "Safe and Adequate Blood for All," primarily focuses on the provision of a safe blood supply [24]. However, the prevalent family-replacement system, which has become a major practice to compel blood donation due to frequent blood shortages at major hospitals in Ghana [27], has engendered a potentially harmful situation, especially at the district level where donated blood is transfused based solely on the screening results from RDTs.

Owing to the scarcity of blood in most Ghanaian blood banks [28], there is a widespread practice where blood is transferred from one facility, often from district health facilities to more advanced hospitals such as regional and teaching hospitals for transfusion to patients, often in critical condition. These units of blood, referred to as "Trans-in blood," are always screened by the transferring facility and only transfused when they test negative for HIV, HBV, HCV, and syphilis, qualifying it for transfusion. It must be emphasized that these units of blood are originally intended for transfusion to patients in the originating facilities but only get transferred when relatives are put under intense pressure to provide blood for their patients who end up being transferred to higher-level facilities for critical care. Given that RDTs are not 100 % specific and sensitive, we hypothesized that at facilities where

blood is screened with only RDTs for transfusion, the risk of transfusing infected blood may be high. This may contribute to thwarting efforts made thus far in combating TTIs in Ghana. Consequently, we investigated the prevalence of missed TTIs in Trans-in donor blood received for transfusion at the Ho Teaching Hospital in the Volta region of Ghana.

Methods

Study design and site

This study utilized a cross-sectional research design to meticulously assess the prevalence of TTIs in blood units obtained from the blood bank of HTH. Originally established as a small-scale health facility in 1999, the HTH transitioned into the Volta Regional Hospital by December 2000 and, following substantial advancements and rigorous accreditation procedures, achieved Teaching Hospital status in April 2019. Presently recognized as the fifth public teaching hospital in Ghana, the institution houses approximately 1200 staff and has over 300 beds across specialized departments including internal medicine, surgery, obstetrics and gyanecology, child health and public health; the hospital also renders blood bank and laboratory services [29,30].

The HTH blood bank is under the leadership of a laboratory manager, technical head and supported by qualified laboratory scientists, donor care nurses, and blood donor organizers. Among its primary functions are the collection of whole blood from eligible donors, thorough screening of donor units for the four TTIs recommended by the WHO [24], blood grouping based on the ABO and RH "D" blood group system and antibody screening, and storage of units or products under standard conditions for transfusion. In response to demand, the blood bank conducts both forward and reverse blood grouping, as well as compatibility testing between donors and potential recipients, adhering to established procedures for transfusion. The principal blood products supplied by the bank include whole blood, Concentrated Red Blood Cells/Packed Red Blood Cells (CRBCs/PRBCs), and Fresh Frozen Plasma (FFP).

Study population and eligibility criteria

The scope of this study encompassed blood units received at the Ho Teaching Hospital blood bank from external healthcare facilities within the period of January 2021 to December 2022. The study only included blood units from source facilities that had undergone prior screening using an RDT method. Data records with missing information, such as the batch number (which serves as a unique identifier for blood units), ELISA screening results, blood group, or the name of the source facility, were excluded from this study.

Sample size determination

The sample size of this study included all trans-in blood units received and associated data documented in the HTH blood bank's trans-in register within the two-year period under review (January 1, 2021, to December 31, 2022).

Processing and transport of trans-in blood units from source facilities

All source facilities from which trans-in blood units were received adhered to a uniform procedure for processing the blood units. At these facilities, potential donors are subjected to a physical examination, a selection interview, and pre-donation testing for TTIs including Hepatitis B, Hepatitis C, HIV, and Syphilis. Additionally, haemoglobin estimation is conducted either using an automated haematology analyzer [31] or the copper sulphate method [32]. Donors who successfully navigate the various selection processes then undergo the bleeding process, during which approximately 350 ml of blood is collected [33]. This blood is stored in a well-labelled blood collection bag and refrigerated at a temperature between 2 and 8 °C [34]. The labels typically include the donor's unique identification number, the date of blood collection, the calculated expiry date, the ABO and Rh "D" blood group, and the results of the TTI screening. The donated blood was transported in iceboxes to maintain the desired cold chain throughout the transport period.

Screening procedures of trans-in blood units at the HTH blood bank

Upon receipt of trans-in units at the HTH blood bank, each blood unit undergoes a comprehensive inspection for haemolysis, leakage, and any indications that may suggest the product is compromised in terms of its quality and safety. Blood units that successfully pass this inspection are accepted for further processing. Subsequently, these units are documented with pertinent information including the potential recipient's name, batch number, collection date, expiry date, blood group, type of product, source facility, contact details of the transporting officer, and the name of the receiving officer. The blood units are then classified according to the ABO and Rh "D" blood grouping system. Following this, they are screened for TTIs using the ChemWell Fusion ELISA & CLIA 4800 Analyzer (Awareness Technologies, United States) testing method [35]. Units that test positive are discarded in accordance with standard operating procedures (SOPs). Conversely, units that test negative are added to the existing inventory and arranged according to their ABO/Rh "D" groups, with consideration given to their expiry dates.

Interpretation of positive and negative ELISA results

In this study, we ascertained the positivity or negativity of the blood units tested using ELISA by comparing the specific absorbances

for each sample with pre-programed reference parameters per the manufacturer's instructions as shown in Table 1. Further, infectivity with any of the TTIs investigated was inferred based on a positive ELISA result. All blood units that tested negative on ELISA were inferred to be non-infected. In all analysis, the selected wavelength (nm) employed was dependent on the instructions provided by the specific reagent manufacturer for each TTI assessed.

Variables and definitions

The dependent variables included in this study are HIV, HBsAg, HCV, and Syphilis whereas the independent variables were ABO, Rh "D" blood groups, and locations where the blood units were transported from.

Data extraction and collection

We apriori developed a bespoke data capture form using Microsoft Excel 2019. This form was designed to extract specific data, including the donor's ABO and Rh "D" blood group, the name of the source facility, the blood unit batch number, and the ELISA screening results for TTIs, specifically HIV, HBsAg, HCV, and Syphilis [24]. The data collection team, comprising staff stationed at the HTH blood bank, utilized this form to meticulously retrieve the necessary records from the blood bank's archives. Records from the trans-in blood archives were cross-referenced with ELISA screening records using the unit batch number, which served as a unique identifier. The data obtained were directly inputted into the Microsoft Office Excel 2019 data capture form.

Spatial visualization of TTIs distribution

The Trans-in blood samples initially lacked the GPS coordinates of their respective source facilities. To address this, the research team made concerted efforts to acquire these coordinates through a combination of telecommunication with accessible facilities, Google Maps, and Ghana Post GPS services. A spatial visualization was subsequently performed to determine the distribution of transfusion-transmissible infections (TTIs) in the Trans-in blood units. Blood units received from each facility were treated as a unique population such that the prevalence of missed TTIs was estimated as a fraction of the total specific TTI positive by the total number of blood units obtained from that facility. Geospatial data visualization techniques, including Geographic Information Systems (GIS), were utilized to create thematic maps. These maps effectively illustrate the prevalence and geographical spread of TTIs within the dataset.

Quality control

In this study, meticulous validation and verification were performed by the data acquisition team to circumvent errors during data entry. This process was executed both before and after data capture into the collection form. Regarding the quality control of the ELISA, the reagent kit encompasses materials for quality control. The assay incorporates both negative and positive controls as integral components of the testing procedure on each run. The legitimacy of the test outcomes is contingent upon the successful passage of these quality controls. In the event of a quality control failure, the test results are considered invalid. Furthermore, routine maintenance is conducted to ensure the apparatus remains in optimal condition, thereby facilitating the production of high-quality results.

Statistical analysis

Data obtained from this study was captured in Microsoft Excel 2019. The data were cleaned by checking for consistency and

Table 1
Interpretation of positive and negative ELISA results.

TTIs	Sample type and storage	Spectral range (nm)	Linearity range	Cut-off value	Positive result interpretation	Negative result interpretation	Borderline results interpretation
HIV	Fresh serum or plasma 2–8 oC	405 450 492 630	−0.2 to 3.0A	NC + 0.12	S/C. $O \geq 1$	S/C. $O < 1$	S/C. $O = 0.9$–1.1
HBV	Fresh serum or plasma 2–8 °C	405 450 492 630	−0.2 to 3.0A	NC x 2.1	S/C. $O \geq 1$	S/C. $O < 1$	S/C. $O = 0.9$–1.0
Syphilis	Fresh serum or plasma 2–8 °C	405 450 492 630	−0.2 to 3.0A	NC + 0.18	S/C. $O \geq 1$	S/C. $O < 1$	S/C. $O = 0.9$–1.0

NC – mean absorbance value for three consecutive controls.
S – sample absorbance.
C.O – Cut-off absorbance.

completeness and then exported to IBM Statistical Package for Social Sciences version 26 (SPSS Inc, Chicago, USA, (https://www.spss.com) for statistical analysis. All data were analyzed descriptively as frequency and percentages and reported in a tabular or graphical form where appropriate. Wilson's confidence interval (CI) was used to determine the 95 % CI of proportions of interest in STATA version 15 (http://www.stata.com) using the STATA command *cii proportions n x, wilson* where *n* is the denominator and *x* is the numerator (frequency of observation of interest). The prevalence of HBV, HIV and Syphilis infection was computed as the ratio of the total number of positive cases of infection to the total number of samples analyzed. Chi-square/ Fisher's exact and Cramer's V or Phi were used to assess the statistical association and magnitude and/or direction of association between variables of interest. Binary logistic regression analysis was used to determine the odds of TTIs based on blood group and location of source facility. Before the spatial visualization, the prevalence of missed TTIs determined for each facility was exported into Microsoft Excel 2019 and merged with obtained individual facility coordinates. The resulting data were then exported to the ArcGIS software where the distance from the individual facilities to the study location was mapped. Further, a colour code was used to represent the distribution and intensity of missed TTIs. P-values of less than 0.05 were considered statistically significant.

Ethical considerations

Ethical approval was obtained from the Ethical and Protocol Review Committee of the Ho Teaching Hospital with protocol identification number **HTH-REC (27) FC 2022**. To ensure data safety and confidentiality, data obtained were stored in a password-encrypted computer folder and accessible only to the researchers.

Results

ABO and Rh "D" blood group characteristics of study participants

This study reviewed a total of 8936 blood bank records between 2021 and 2022. Of this, 727 units of blood, 8.14 % (95 % CI: 7.59–8.72) were transferred from various facilities to the Ho Teaching Hospital (trans-in). We observed the traditional ABO blood group 'O' predominance (68.91 %) among the blood donors in this study, followed by group B (15.82 %) and group A (14.17 %). Also, Rh "D" Positive predominated (82.53 %) among the donors in this study (Table 2).

Prevalence of missed TTIs in trans-in blood received at the Ho teaching hospital between 2021 and 2022

The overall prevalence of missed TTI was 6.19 % (95 % CI: 4.66–8.18) in this study and this comprised missed HIV, HBV, and Syphilis infection rates of 1.79 % (95 % CI: 1.72–1.86), 1.24 % (95 % CI: 1.17–1.31), and 3.16 % (95 % CI: 3.09–3.23), respectively. As depicted in Table 3, the prevalence of TTI was highest among donors with ABO blood group "O", 68.89 (95 % CI: 54.34–80.47) and Rh "D" Positive blood group, 95.56 % (95 % CI: 85.17–98.77). However, the highest missed HIV, HBV, and Syphilis infections was observed among donors with ABO blood group "O", 2.40 % (95 % CI: 2.31–2.49), "B", 3.48 % (95 % CI: 3.30–3.66) and "A", 5.83 % (95 % CI: 5.64–6.02) respectively. When stratified by Rh "D" status, we observed the highest missed HIV, HBV, and Syphilis infections of 2.00 % (95 % CI: 1.92–2.08), 1.50 % (95 % CI: 1.42–1.58), and 3.67 % (95 % CI: 3.59–3.75), respectively among donors with Rh "D" Positive blood group. Based on location, HIV, 2.00(95 % CI: 1.15–3.46), HBV, 1.50(0.79–2.82) and Syphilis, 3.33(2.16–5.08) infections were highest among blood units received from facilities outside Ho as against those within Ho. There was however no significant association between ABO blood group, Rh "D" status and location and the prevalence of missed TTIs.

Yearly burden of TTIs in trans-in blood received at the Ho teaching hospital between 2021 and 2022

The burden of missed TTIs for 2021 and 2022 were statistically comparable ($p > 0.05$). However, except for HIV infection where the rates of missed infections were higher in 2022, 2.04 % (95 % CI: 1.04–3.97) compared to 2021, 1.49 % (95 % CI: 0.64–3.45), there was

Table 2
ABO and Rh "D" blood group characteristics of study participants.

Blood Group	Total population n (%)
Overall	727(8.14)
ABO Blood Group	
Blood Group "O"	501(68.91)
Blood Group "A"	103(14.17)
Blood Group "B"	115(15.82)
Blood Group "AB"	8(1.10)
Rh "D" Blood Group	
Rh "D" Negative	127(17.47)
Rh "D" Positive	600(82.53)
Location	
Within Ho	126(17.33)
Outside Ho	601(82.67)

Table 3
Prevalence of missed TTIs in trans-in blood received at the Ho Teaching Hospital between 2021 and 2022.

Blood Group	Total population n [% (95 % CI)]	HIV n [% (95 % CI)]	HBV n [% (95 % CI)]	Syphilis n [% (95 % CI)]
Overall	45[6.19(4.66–8.18)]	13[1.79(1.72–1.86)]	9[1.24(1.17–1.31)]	23[3.16(3.09–3.23)]
ABO Blood Group				
Blood Group "O"	31[68.89(54.34–80.47)]	12 [2.40 (2.31–2.49)]	3[0.60(0.51–0.69)]	16[3.19(3.10–3.28)]
Blood Group "A"	9[20.0(10.90–33.82)]	1[0.97(0.78–1.16)]	2[1.94(1.75–2.13)]	6[5.83(5.64–6.02)]
Blood Group "B"	5[11.11(4.84–23.50)]	0(0.00)	4[3.48(3.30–3.66)]	1[0.87(0.69–1.05)]
Blood Group "AB"	0(0.0)	0(0.00)	0(0.00)	0(0.00)
P-value		0.298	0.705	0.202
Cramer's V		0.071	0.097	0.080
Rh "D" Blood Group				
Rh "D" Negative	2[4.44(1.23–14.83)]	1[0.79(0.62–0.96)]	0(0.00)	1[0.79(0.62–0.96)]
Rh "D" Positive	43[95.56(85.17–98.77)]	12[2.00(1.92–2.08)]	9[1.50(1.42–1.58)]	22[3.67(3.59–3.75)]
P-value		0.710	0.372	0.156
Phi		0.035	0.052	0.062
Location				
Within Ho	4[8.89(3.51–20.73)]	1[0.79(0.14–4.36)]	0(0.00)	3[2.38(0.81–6.77)]
Outside Ho	41[91.11(79.27–96.49)]	12[2.00(1.15–3.46)]	9[1.50(0.79–2.82)]	20[3.33(2.16–5.08)]
P-value		0.354	0.167	0.581
Phi		0.034	0.051	0.021

a decline in the rates of missed HBV and Syphilis infection from 2021 (3.28 %) to 2022 (3.06 %) (Phi = −0.046 for missed HBV; Phi = −0.006 for missed Syphilis infection) (Fig. 1).

Coinfection rates of TTIs in blood units received at the Ho teaching hospital between 2021 and 2022

Further analysis revealed at least two units presenting with missed HIV and HBV as well as missed HIV and syphilis co-infection rates each of 0.28 % (95 % CI: 0.08–0.10). However, three units presented with missed HBV and syphilis co-infection rate of 0.41 % (95 % CI: 0.14–1.21) (Fig. 2).

Odds ratios of HIV, HBV, and syphilis across ABO and Rh "D" blood groups

In this study, it was found that the odds of missed HIV and syphilis infections were lower by 82 % (95 % CI: 0.02–1.40) and 3 % (95 % CI: 0.39–2.39) respectively for donors with non-O blood groups compared to those with blood group "O", although these findings were not statistically significant. On the other hand, the odds of missed HBV infection was 4.53 times higher (95 % CI: 1.12–18.27; p-value=0.034) for non-O blood group donors in comparison to blood group "O" donors. Moreover, donors with Rh "D" Positive blood groups showed higher odds of missed HIV (OR=2.57, 95 % CI:0.33–19.96) and syphilis (OR=4.80, 95 % CI: 0.64–35.91) infections, but these results were not statistically significant. Similarly, compared to source facilities within Ho where the receiving facility is located,

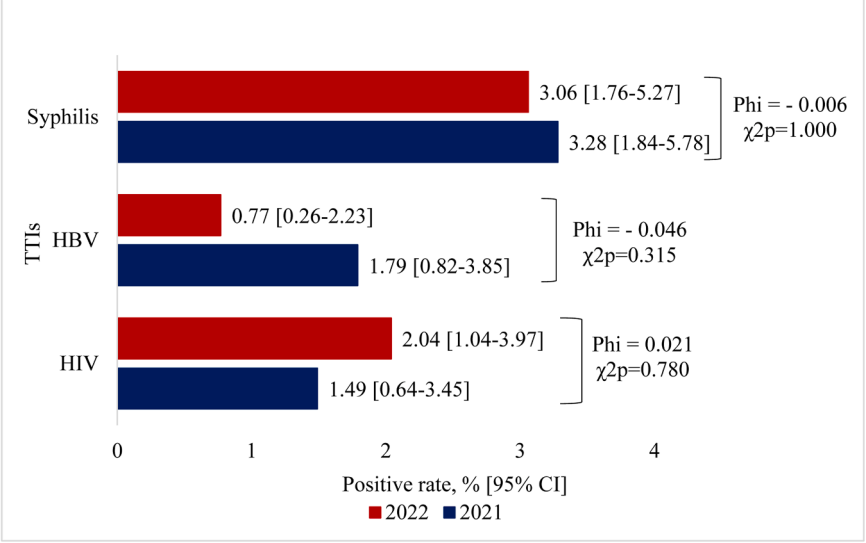


Fig. 1. Yearly burden of TTIs in trans-in blood received at the Ho Teaching Hospital between 2021 and 2022. χ2p-Chi-square p-value.

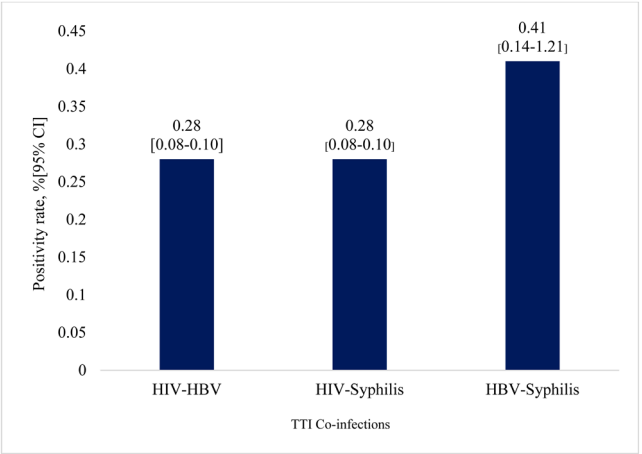


Fig. 2. Coinfection rates of TTIs in blood units received at the Ho Teaching Hospital between 2021 and 2022.

source facilities outside of Ho showed 2.55 and 41 % higher odds of missed HIV (95 % CI: 0.33–19.76), and missed Syphilis (95 % CI: 0.41–4.82) infections although not significant (Table 4).

A distance map of health facilities from HTH where blood units were received

The map in Fig. 3 below describes the distance of the various health facilities from which blood units were received for transfusion at the Ho Teaching Hospital. Most of the facilities were close to the Ho Teaching Hospital and with a proximity of 92 km away. A moderate number of the facilities were between a distance of 92 km and 181 km from HTH. Only three of the facilities were further than 181 km from the hospital but none of the facilities were further the 271 km away from HTH.

Spatial distribution of missed HBV in trans-in blood units at HTH between 2021 and 2022

We observed in this study that the burden of HBV in the trans-in blood units assessed to be more concentrated in the south-eastern part of the Volta region of the country particularly in areas around Aflao and Keta (Fig. 4).

Spatial distribution of missed syphilis infection in trans-in blood at HTH between 2021 and 2022

We also observed as shown in Fig. 5 that missed cases of syphilis were densely distributed in areas around the northern part of the Volta Region and in the Oti Region, around the areas of Kpando, Hohoe, Jasikan, Nkwanta and adjacent communities with proportions ranging from 0.14 % to 0.42 %. Additionally, there was a cluster of syphilis burden in blood units received from areas in the south-eastern part of the Volta Region spanning Aflao and its environs.

Spatial distribution of missed syphilis infection in trans-in blood at HTH between 2021 and 2022

Spatial distribution of missed HIV infection in trans-in blood at HTH between 2021 and 2022

As depicted in Fig. 6, we observed a 0.14 % burden of missed HIV cases evenly distributed throughout the Volta Region with a

Table 4
Odds Ratios of HIV, HBV, and Syphilis Across Blood Groups and location of source facility.

Blood Group	Missed HIV OR [95 % CI]	Missed HBV OR [95 % CI]	Missed Syphilis OR [95 % CI]
ABO Blood Group			
Blood Group "O"	1	1	1
Blood Group "non-O"	0.18 [0.02–1.40]	4.53 [1.12–18.27] *	0.97 [0.39–2.39]
Rh "D" Blood Group			
Rh "D" Negative	1	1	1
Rh "D" Positive	2.57 [0.33–19.96]	1 #	4.80 [0.64–35.91]
Location of source facility			
Within Ho	1	1	1
Outside Ho	2.55 [0.33–19.76]	1#	1.41 [0.41–4.82]

Omitted due to perfect failure prediction.
* P-value significant at 0.05.

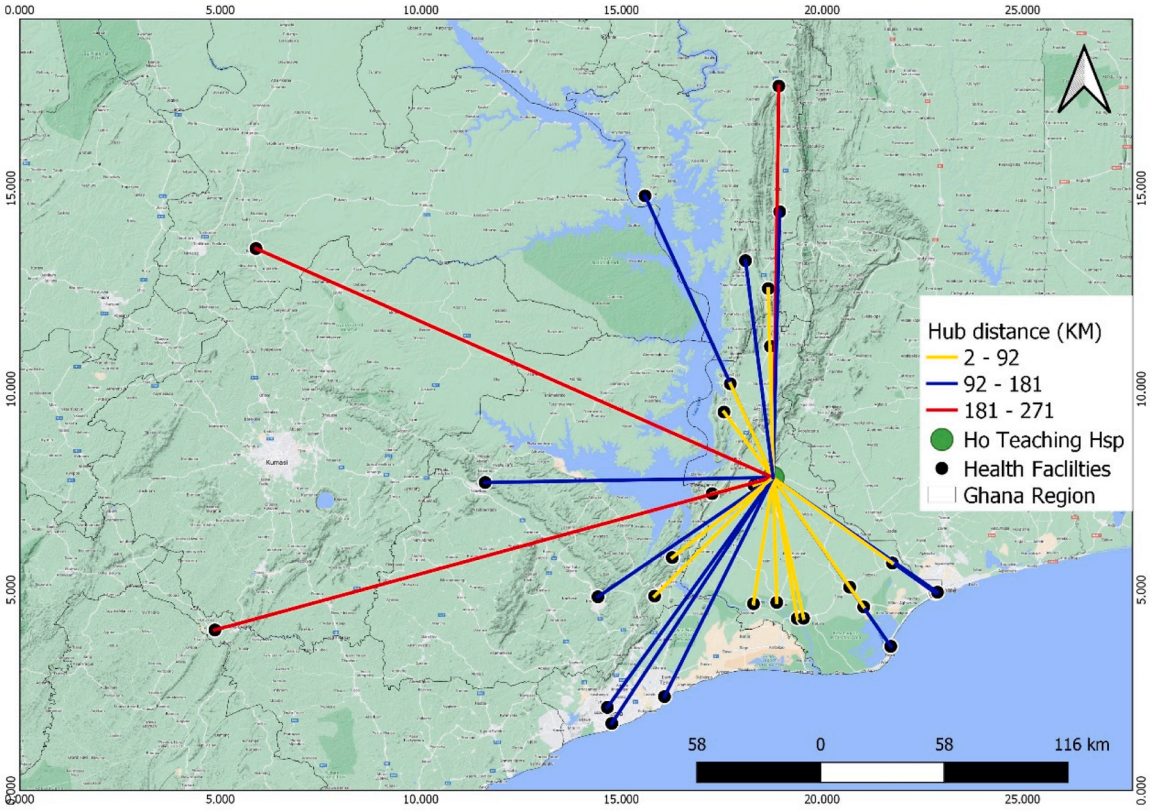


Fig. 3. A distance map of health facilities from HTH where blood units were received.

cluster of the infection (0.28 % prevalence) noted around areas in the south-eastern part of the Volta Region spanning Aflao and Keta.

Discussion

This study investigated 727 trans-in blood units, representing 8.14 % (95 % CI: 7.59–8.72) of the total 8936 archived results at the Ho Teaching Hospital for the two years under review. The overall prevalence of missed TTI was 6.19 % (95 % CI: 4.66–8.18), comprising 1.79 % (95 % CI: 1.72–1.86) of missed HIV, 1.24 % (95 % CI: 1.17–1.31) of missed HBV, and 3.16 % (95 % CI: 3.09–3.23) of missed Syphilis infections. The prevalence of missed TTI was most pronounced among donors with the ABO blood group "O", 68.89 % (95 % CI: 54.34–80.47), and the Rh "D" positive blood group, 95.56 % (95 % CI: 85.17–98.77). The results showed higher missed HIV infection prevalence in 2022 (2.04 %, 95 % CI: 1.04–3.97) compared to 2021 (1.49 %, 95 % CI: 0.64–3.45), but a decline in the rates of missed HBV and Syphilis infections from 2021 to 2022 was observed. The highest missed rates of HIV, HBV, and Syphilis infections were observed among donors with the ABO blood groups "O" (2.40 %, 95 % CI: 2.31–2.49), "B" (3.48 %, 95 % CI: 3.30–3.66), and, and "A" (5.83 %, 95 % CI: 5.64–6.02). However, when stratified by Rh "D" status, the highest missed rates of HIV, HBV, and Syphilis infections of 2.00 % (95 % CI: 1.92–2.08), 1.50 % (95 % CI: 1.42–1.58), and 3.67 % (95 % CI: 3.59–3.75) respectively, were observed among donors with the Rh "D" positive blood group. Similarly, HIV, 2.00(95 % CI: 1.15–3.46), HBV, 1.50(0.79–2.82) and Syphilis, 3.33 (2.16–5.08) prevalence rates were highest among blood units received from facilities outside Ho as against those within Ho. At least two units presented with missed co-infections of HIV and HBV, as well as HIV and Syphilis, each at a rate of 0.28 % (95 % CI: 0.08–0.10), whereas three units presented with a missed co-infection rate of HBV and Syphilis of 0.41 % (95 % CI: 0.14–1.21). There was no significant association between the ABO blood group, Rh "D" status and location and the prevalence of missed TTIs however, the odds of missed HBV infection was 4.53 times higher (95 % CI: 1.12–18.27; p-value=0.034) for non-O blood group donors in comparison to blood group “O” donors.

The overall prevalence of missed TTIs identified in this study is lower than the 10.1 %, 14.96 %, 14.1 %, and 21.0 % TTI prevalence rates reported in Tanzania [36], Calabar, Nigeria [37], Kenya [38] and Ghana [39] respectively but however higher than the 2.0 % reported in Southern Africa [40]. While we recognize the similarities in disease components between the studies, we exercise caution in comparing the prevalence rates of missed TTIs as observed in our study to the TTIs reported in the general population, as this may lead to interpretation inaccuracies. It must be emphasized that the prevalence of TTIs documented in our study was from blood units that were declared pathogen-free for transfusion at the originating facilities. Given the rate observed in this study and considering the devastation these infections cause to livelihoods, this could raise issues of public health concern and should be given the necessary



Fig. 4. Spatial distribution of missed HBV in trans-in blood units at HTH between 2021 and 2022. HBV occurrence is presented as a percentage (%).

attention. Looking ahead, improvements in the screening methods of donor blood, such as the use of individual donor-nucleic acid testing (ID-NAT), should be prioritized by the National Blood Service and the relevant stakeholders. Although ID-NAT has been reported by Vermeulen [41] to miss the detection of HBV in some instances, the technique has been shown to substantially improve blood safety for both HIV and HBV and also proved beneficial in the detection of donors with occult blood infections creating risk for HBV [40].

The overall prevalence of missed HIV infection (1.79 %) observed in this study is higher compared to the 1.0 % reported by the WHO [40] and the 1.66 % national prevalence documented by the Ghana AIDS Commision [42]. With an antiretroviral therapy coverage rate of only 64.1 % among adults aged 15 years and above, and 43.6 % among children aged 0–14 years [42], the impact of missed HIV detection could be detrimental and overwhelming if not addressed with the appropriate policy mechanisms. Additionally, the financial implications that may arise following the exacerbation of the HIV epidemic promise to be severe, given that Ghana continues to default on its co-funding financial obligations to the Global Fund for HIV/AIDS support. As a result, for the period 2021 to 2025, even with donor support, Ghana is facing a funding gap of US$445 m, representing approximately 67 % of the total cost of the National HIV & AIDS Programme [43]. This could pose serious challenges to HIV prevention and treatment strategies in Ghana in the near future. While we acknowledge the array of interventions that have been employed in the prevention of HIV transmission and treatment regimens [44,45], we strongly suggest that a shift towards addressing the flaws in screening and/or diagnostic methods, especially at the district levels, will undoubtedly be essential in contributing to the fight against the HIV epidemic in Ghana.

In comparison to the estimated national HBV prevalence of 8.36 % in Ghana [46] and HBV prevalence reported among blood donors in various studies [3,39,40,47], the overall missed HBV prevalence of 1.24 % documented in this study is considerably low. However, given that individuals requiring blood transfusion are often immunocompromised or immunodeficient, such as patients with inherited blood disorders, pregnancy-related complications, chronic diseases causing anaemia, etc., the impact of HBV infection among these populations may worsen treatment outcomes. In addition to these, the risk of initiating a transmission chain, such as vertical transmission from mother to child, may be high. It's crucial to note that neonates born to mothers infected with chronic HBV have a 90 % risk of acquiring the chronic form of the infection [48], emphasizing the need to ensure the safety of blood before transfusion. With the current focus on testing and treatment [49], special attention must also be paid to prevention at all levels, particularly through haemotherapy, where the risk spans all age groups, demographics, and people.

In our study, we observed a relatively high syphilis prevalence of 3.16 %, compared to what is typically seen in the general blood donor population, including 0.70 % in Western China [50], 0.87 % in Brazil [51], 0.02 % in Amman, Jordan [52], 2.1 % in Pakistan

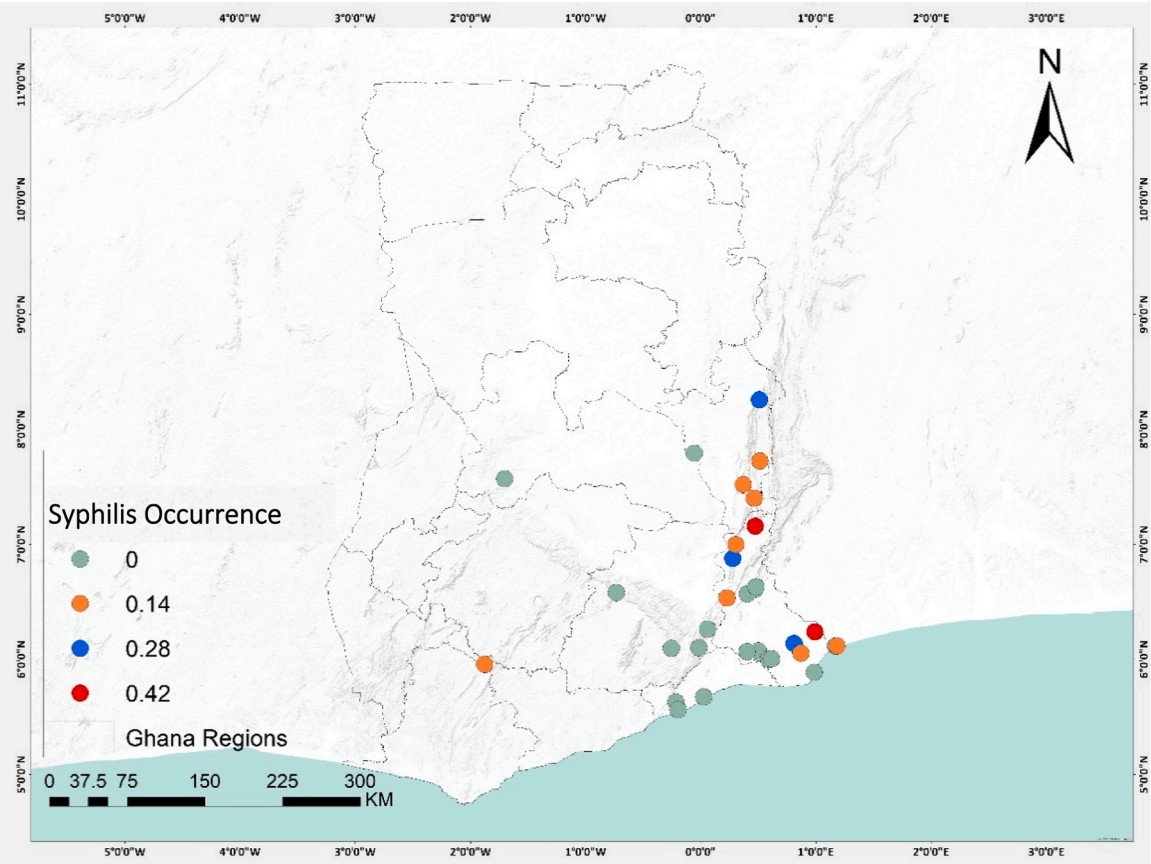


Fig. 5. Spatial distribution of missed syphilis infection in trans-in blood at HTH between 2021 and 2022. Syphilis occurrence is presented as a percentage (%).

[53] and 0.9 % in Ethiopia [47]. However, a few studies reported higher prevalence rates ranging between 3.3 % to 6.8 % [3,39,54]. The current missed syphilis prevalence observed in our study is also higher than that documented by both a population-based survey (2.58 %) and a Sentinel Survey (0.24 %) in Ghana [55] raising serious public health concerns for haemotherapy in Ghana. Although we lack comprehensive data on donor demographic characteristics to determine the primary sex and age group from which the various blood units originated, we conjecture that considering the general eligibility criteria for blood donation and the fact that more males than females, especially in Ghana, are found to be involved in blood donation [3], increased sexual debut characterized by high promiscuity among males may be the major contributing factor for this finding [3]. It has therefore become paramount to ensure that more sensitive and robust screening methods are prioritized to significantly reduce the risk of TTIs to patients who need blood for care and health restoration.

The epidemics of HBV and HIV have been at the forefront of infectious disease discourse in Ghana, often overshadowing other infections such as syphilis. This disproportionate focus may account for the higher prevalence of missed syphilis infections observed in this study, which exceeded those of HIV and HBV in both 2021 (3.28 %) and 2022 (3.06 %). Presumably, this trend may be a reflection of observed global patterns, as evidenced by the documented escalation in syphilis cases within the demographic of men who have sex with men (MSMs) and congenital syphilis cases [56]. Given these findings, it is crucial to advocate for equal attention to all four TTIs. This is a conversation that must be initiated by relevant stakeholders, because a setback in combating one infection can undermine the overall fight against all. Meanwhile, the slight decrease in syphilis prevalence from 2021 to 2022 aligns with the trend reported by Osei-Yeboah [55] between 2012 and 2016. Despite syphilis being treatable, the health consequences for infected individuals can be significant [56] if the infection is not detected and treated promptly. As a result, more robust and sensitive testing methods should be considered for screening donor blood. On the other hand, the observed trend in missed HIV infections over the two years under study from 1.49 % in 2021 to 2.04 % in 2022 is concerning. This rise could be attributed to the reported increase in the incidence of HIV in Ghana, which escalated from 19,000 new infections in 2020 to 23,495 new infections in 2022 [44,45]. Given the unique characteristics of our study population, it is imperative to enhance the testing methods used in the screening of donor blood for transfusion. Without such improvements, the battle against TTIs may remain unresolved, potentially leading to a persistent, if not worsening, burden. Furthermore, while the prevalence of co-infections seen in this study is relatively low, it could significantly increase the burden of the infection if the current rate of missed detection persists.

In this study, we observed that missed TTIs were primarily localized in the regions surrounding Ho and Aflao. Specifically, missed

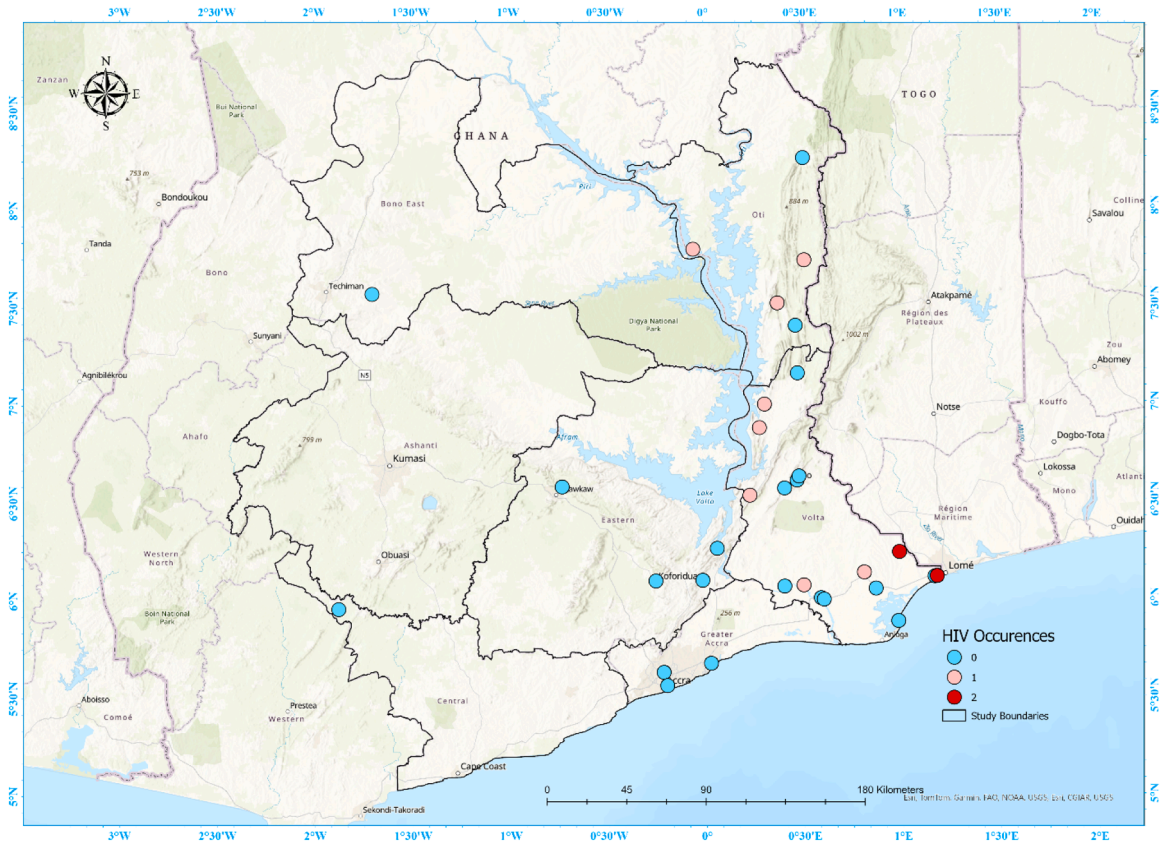


Fig. 6. Spatial distribution of missed HIV infection in trans-in blood at HTH between 2021 and 2022. HIV occurrence is presented as a percentage (%).

Hepatitis B Virus (HBV) was concentrated in the south-eastern part of the Volta region, particularly around Aflao and Keta (Fig. 4). In contrast, missed syphilis cases (Fig. 5) were densely distributed in the northern part of the Volta Region and in the Oti Region, specifically around Kpando, Hohoe, Jasikan, Nkwanta, and neighbouring communities, as well as in the south-eastern part of the Volta Region, encompassing Aflao and its surroundings, with proportions ranging from 0.14 % to 0.42 %. The literature is currently limited regarding the geographical distribution of missed TTIs in Ghana, and this restricted our ability to make in-depth comparisons between our findings and those of other studies. With the data currently available, it is uncertain whether the observed trend of missed HBV and syphilis prevalence represents a unique pattern. However, several factors could potentially explain this finding. Firstly, the observed distribution of missed syphilis could be attributed to the fact that these locations are densely populated areas with high economic activity, which may foster behaviours that increase the risk of syphilis transmission. Furthermore, these areas are primarily border locations, either between countries (as in the case of Aflao, which leads to Togo) or between regions (as in the case of Kpando and Hohoe in the Volta region leading to Jasikan in the Oti region, and then from Nkwanta in the Oti region to the Northern region of Ghana). These border towns are typically characterized by complex demographics of individuals who may engage in activities that could facilitate the contraction and subsequent spread of syphilis in the population. These population dynamics may have contributed to the distribution of missed syphilis observed in this study at these locations. However, the focal concentration of missed HIV prevalence observed in the south-eastern part of the Volta region, according to our findings, could be attributed to the relatively high HIV prevalence among the general population in those communities, specifically 1.92 % in Keta, 1.15 % in Ketu North, and 1.37 % in Ketu South [42]. Further studies are necessary to fully understand the influence of population dynamics and geographic locations on missed TTIs in Ghana. This knowledge will inform policy on the proper interventions needed to curb the spread of these TTIs and consequently make blood safer for transfusion.

Our results showed that the highest missed HIV, HBV, and syphilis infections were among donors with ABO blood group "O" (2.40 %), "B" (3.48 %) and "A" (5.83 %) respectively and among donors with Rh "D" positive blood group (2.00% vs 1.50% vs 3.67 % respectively). The odds of missed HBV infection were 4.53 times higher (95 % CI: 1.12–18.27; p-value=0.034) for non-O blood group donors in comparison to blood group "O" donors. In an akin study, the highest HIV, HBV and syphilis prevalence were recorded among donors with blood group "B-" (0.07 %) and "*B*+" (0.07 %), "AB-" (0.41 %) and B- (0.33 %) respectively [57]. Elsewhere, blood group "A-" (1.35 %), "A-" (2.70 %) and "AB+" (0.27 %) recorded the highest HIV, HBV and syphilis prevalence rates among donors [58]. The role played by ABO and Rh "D" blood groups in the risk of TTIs and any other pathogens for that matter has not been fully explored. While some studies have reported no significant association between HIV, HBV, HCV and syphilis TTIs with ABO and Rh "D" blood groups

[59,60], other studies found significant relationship between at least one transfusion transmissible pathogen and ABO and/or Rh "D" blood groups [57,61,62]. Although blood groups, specifically the ABO blood group have been shown to influence vulnerability to diseases via methods such as serving as a gateway for pathogens and controlling immune reactions through antibody production [63, 64], not much is known about the causal influence of blood group on TTIs. It is however important to emphasize that with such a high prevalence of missed superbugs among the ABO blood groups and most importantly, among donors with Rh "D" positive, stringent measures need to be instituted to curb the further spread of these pathogens within the population. The detrimental effect including a reduction in the already hard-to-find donor population in Ghana cannot be underestimated.

Study limitation

While this study provides valuable insights, certain limitations are worthy of note. The missed positivity rate of syphilis was not validated by a confirmatory assay; thus, the observed rate serves only as an indicative marker. The higher rates in our study could be explained by the detection of syphilis specific antibody by the ELISA which is known to remain present and test positive several years after resolution of infection as opposed to the non-treponemal test such as VDRL and RPR which may have been used for screening by the district hospitals [54,65,66]. Also, although ELISA tests are typically recognized for their sensitivity and specificity, their precision can fluctuate. Specifically, when detecting HBV and HIV, ELISA tests may exhibit lower accuracy and a higher incidence of false negatives in comparison to other methodologies which may affect the reliability of the results in this study. The study's data covers only two years (2021 and 2022), and a longer duration might offer a more comprehensive understanding of the prevalence of missed TTIs and their public health impact. The lack of demographic data such as age, sex, occupation, and lifestyle habits of the donors may limit the identification of infection-associated risk factors. Furthermore, due to its retrospective nature, the study could not control or provide extensive information on the risk factors for TTIs, which might have influenced the results.

Conclusion

This study reveals a high incidence of missed TTIs in blood received at the Ho Teaching Hospital, particularly among donors with 'O' and Rh "D" Positive blood groups. The findings underscore potential shortcomings in current screening procedures, raising significant public health concerns. Considering that the trans-in blood units were declared safe for transfusion at the source facilities, there was a clinical risk that recipients of these blood units could develop infections from the transfusions. Therefore, to the extent that the trans-in blood units analyzed in this study were a fraction of the units screened and cleared for transfusion at the source facilities, the findings reported in this study could be an underestimation of the true picture in the population, as many more patients stand the risk of being exposed to these TTIs. Thus, if the trend of missed TTIs persists, it could undermine efforts to control the spread of these infections within the population in Ghana.

Recommendations

We recommend that future studies employing more robust techniques should focus on elucidating other deficiencies in the existing screening protocols or tools and encourage improvements in the quality of testing or screening of TTIs, especially in blood units meant for haemotherapy. Further, studies among larger populations should be conducted to ascertain the burden of the missed infections in Ghana to inform policy decisions. In the meantime, it is crucial to enhance the sensitivity of blood screening methods to ensure transfusion safety.

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Data availability

The data used for the study have been included in the manuscript.

CRediT authorship contribution statement

Renosten Edem Tetteh: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Evelyn Akosua Yeboaa:** Data curation, Writing – original draft, Writing – review & editing. **Wisdom Yayra Morganu-Dogbey:** Data curation, Writing – original draft, Writing – review & editing. **Emmanuel Ativi:** Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Francis Bigoja:** Data curation, Writing – original draft, Writing – review & editing. **Elom Yawa Dzefi:** Writing – review & editing. **Patrick Komla Affrim:** Supervision, Writing – review & editing. **Prosper Mensah:** Supervision, Writing – review & editing. **Precious Kwablah Kwadzokpui:** Conceptualization, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Ethical Approval Letter



In case of reply the number
And the date of this
Letter should be quoted
My Ref. No. HTH/
Your Ref. No.
Our Core Values:
** Commitment
** Accountability
** Dedication
** Integrity
** Professionalism
** Innovation
** Teamwork
** Safe Care

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Website: www.hth.gov.gh

2nd January, 2023

ETHICAL APPROVAL

Principal Investigator: **Renosten Edem Tetteh**

Protocol ID NO: **HTH-REC (27) FC_2022**

Protocol Title: **"Assessing the performance of RDTs in screening for Transfusion Transmissible Infections (TTI) among blood donors and the occurrence of missed infections."**

The Ho Teaching Hospital Research Ethics Committee upon considering the ethical merits has approved your proposal. This approval requires that you fulfil the following conditions.

- Submit periodic progress report during field work and submit final or study closure report to the HTH-REC.
- The HTH-REC may perform periodic monitoring and evaluation to ensure compliance with the protocol as approved.
- You are to report adverse event related to this study verbally within one week and in writing within two weeks.
- Any significant protocol amendment must be resubmitted to the committee for approval before implementation.
- You are required to notify the committee before publishing any research finding related to this study.

This approval is valid until **1st January, 2024** after which you have to apply for renewal. Please quote protocol identification number in future correspondence related to this protocol.

Rev. Dr. S.T.K. Dzokoto
Chairman
Ho Teaching Hospital Research Ethics Committee (HTH-REC)

Research Ethics Committee (REC)
Ho Teaching Hospital,
P.O. Box MA 374, HO, Ghana
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